COLOURS AND VISION

PYQ • 2021 - 2026

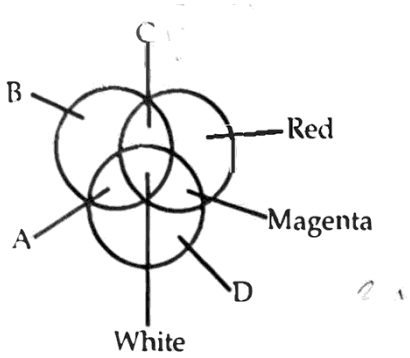
1. 2026 **3 MARKS**

Observe the figure where three coloured lights are made to fall on a screen.

(a) Which colours are labelled as A and D ?

(b) Among the colours A, B, C and D identify the primary colours and write down the colours.

(c) There is a complementary colour pair among A, B, C and D. Pick them out and write down.



2. 2026 **2 MARKS**

(A) The Sun appears red or yellow or orange during sunrise and sunset. But normally the sky appears in blue colour. Describe an experiment that would be helpful in explaining the cause of these phenomena.

OR

(B) In winter, path of light through the gaps of the branches of trees can be seen clearly due to scattering.

(a) By what name is the phenomenon known where the path of light becomes visible in this manner ? Through what type of medium does light travel when this occurs ?

(b) What does the intensity of scattering depend on ?

3. 2026 **1 MARK**

When passing through water droplets while forming rainbow light rays undergo

(i) Increase in speed. (ii) Internal reflection. (iii) Total internal reflection. (iv) Refraction.

(a) (i) and (ii) are correct (b) (ii) and (iii) are correct

(c) (i), (ii) and (iv) are correct (d) (ii) and (iv) are correct

4. 2025 **4 MARKS**

The irregular and partial reflection of light is called scattering.

(a) Write the factors that influence scattering.

(b) Explain the reason for the red colour of the setting sun ?

(c) The sky always appears dark for an astronaut in space. Why ?

5. 2025 **3 MARKS**

Observe the following diagram regarding the image formation in the eye.

(a) Identify the diagram that shows long-sightedness (hypermetropia). Justify your answer.

(b) How can this defect be corrected ?

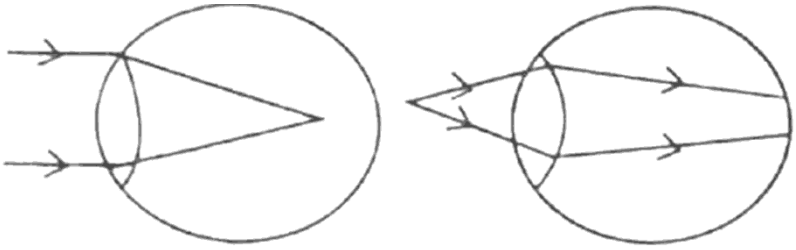


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6. 2024 **4 MARKS**

Give reason for each of the following.

(a) A rapidly rotating Newton's colour disc appears white.

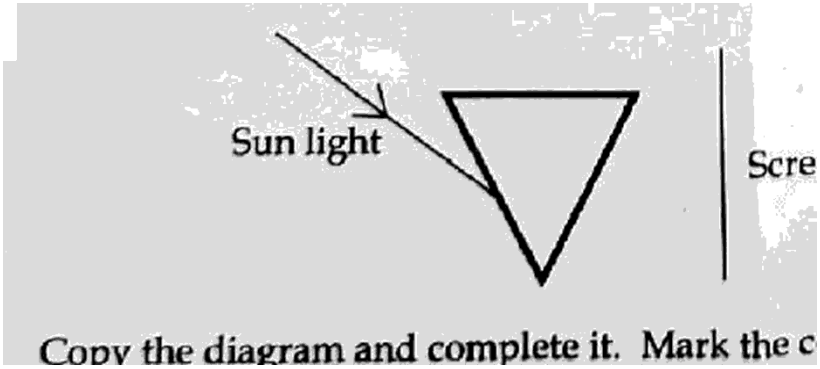
(b) Sun appears red during sunrise and sunset.

(c) A clear sky appears blue.

(d) On frosty mornings the path of sunlight passing through the woods is clearly visible.

7. 2024 **2 MARKS**

The figure shows sunlight falling obliquely on a glass prism. Copy the diagram and complete it. Mark the colours seen on the screen.



8. 2024 **1 MARK**

If the near point of a person is 25 cm and the far point is not infinity then his eye has _____ .

(No defect, Myopia, Hypermetropia, Presbyopia)

9. 2023 **4 MARKS**

The red colour during sunset and the beautiful rainbow make nature colourful.

(a) Name the phenomenon that creates a rainbow.

(b) What is the colour seen at the upper edge of the primary rainbow ?

(c) During sunset sun seems to be in red colour. Explain why ?

10. 2023 **3 MARKS**

A doctor's prescription for a person with defective vision have the figures -1D, -1.25D.

(a) What do these figures indicate ?

(b) Identify the defect of the eye.

(c) Write the reason for this defect.

11. 2023 **1 MARK**

What is the distance to the near point from an eye with healthy vision ?

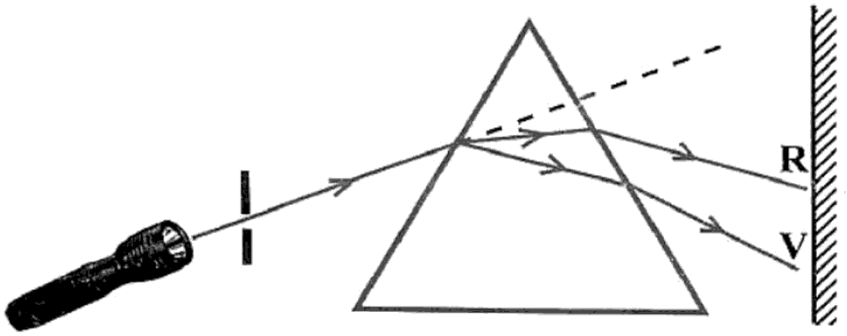
12. 2022 **3 MARKS**

The figure given below shows dispersion of white light when it passes through a prism.

(a) Which colour deviates more ?

(b) Which colour of the visible light has the longest wavelength ?

(c) During dispersion each colour has got different deviations. Why ?



13. 2022 **2 MARKS**

Why does Newton's colour disc appear to be white, when it is rotated at high speed ? Explain.

14. 2022 **1 MARK**

When light passes through a medium it suffers partial and irregular reflection by hitting the particles of the medium. Name this phenomenon.

15. 2021 **3 MARKS**

Rainbow is formed due to dispersion of sunlight.

(a) What is dispersion ?

(b) Write down the changes in the sunlight in the water droplet as the rainbow forms. Explain.

16. 2021 **1 MARK**

Select the odd one from the group.

[Reflection, Dispersion, Refraction, Persistence of vision]

17. 2021 **1 MARK**

Choose the least scattered colour in sunlight from the following.

[Violet, Green, Blue, Red]